BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 099 · Lesson 099 · CISC and RISC: Instruction Set Design

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is ISA?

Q2. What does an opcode identify?

Q3. What is a load/store instruction?

Q4. Simplify $(A \wedge B) \vee (A \wedge \neg B)$.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

ISA — Programmer-visible machine instruction set.

CISC — Complex Instruction Set Computer.

RISC — Reduced Instruction Set Computer.

load/store — Explicit memory-register data transfer.

New learning

Observe — CISC typically offers a larger/more complex instruction repertoire, with instructions able to combine operations; RISC generally uses more regular, often simple operations.

Apply — RISC load/store designs explicitly move operands between memory and registers and perform ALU work using registers.

Evaluate — Avoid outdated absolutes: modern CISC and RISC microarchitectures can both pipeline, use caches and achieve strong performance; trade-offs depend on implementation.

Worked example

Model a RISC load-add-store sequence for A5,B8 and result C.

1. Follow the reasoning — LOAD R1,A→5; LOAD R2,B→8; ADD R3,R1,R2→13; STORE C,R3→13.

We Do · Practise with guidance

For A4,B7, trace the same abstract load/store sequence and state result.

Follow the reasoning

R1=4,R2=7,R3=11, memory[C]=11.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

State two common RISC and two common CISC characteristics. [4]

Question 2

Trace the abstract RISC sequence above when A=6,B=9, give R1,R2,R3,C. [6]

Question 3

Evaluate statement 'CISC processors cannot use pipelining and RISC always uses fewer instructions'. [6]

Exit Ticket · Answer before checking

Modern CISC implementations cannot use pipeline stages.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Instruction Set Architecture—programmer-visible instruction operations.

Q2. Which operation the CPU performs.

Q3. Transfers values between registers and memory.

Q4. $A \wedge (B \vee \neg B) = A$.

You Do pages · Independent questions

Q1. RISC regular/simple instructions, load/store architecture [2]; CISC richer/variable instructions, potentially combined work [2].

Q2. R1=6 [1], R2=9 [1], R3=15 [2], C=15 [2].

Q3. Both can pipeline [2]; RISC often needs more simple ISA instructions for complex task [2]; real microarchitecture and compiler/workload matter [2].

Exit Ticket · Model answer

False. CISC processors use pipelines too.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: